

Home Lab 2: The Powder Workflow

Do this with a parent. Goggles on. Never taste anything. Clean up after. See the [Home Lab Safety Agreement](#).

Goal

Practice running the same tests in the same order on four known powders, so it becomes automatic.

You need

From the kit: **baking soda, corn starch, table salt, sugar**; iodine dropper; cabbage indicator.
From home: white vinegar, water, a muffin tin or 4 cups, toothpicks, paper towels.

Steps

For **each** of the four powders, in this order:

1. **Look** — write down what it looks like (crystal? fine powder?).
2. **Feel** — rub a pinch between your fingers. Gritty? Silky? Soft?
3. **Water** — stir a pinch into a little water. Does it fully dissolve, stay cloudy, or make "oobleck"?
4. **pH** — add 2 drops of cabbage indicator to the water mix. Color? Acid, neutral, or base?
5. **Acid** — put a fresh pinch on a paper towel, add a few drops of vinegar. Fizz or no fizz?
6. **Iodine** — put a fresh **dry** pinch on the paper towel, add **one** drop of iodine. Blue-black, or stays orange-brown?

Do all four. Then do it a second time, faster, and see if you can do it from memory without checking the steps.

Results

Powder	Look	Feel	In water	pH (cabbage)	+ Vinegar	+ Iodine
Baking soda						
Corn starch						
Table salt						
Sugar						

Follow-up

1. Which powder fizzed with vinegar? What gas made the fizz?
2. Which powder turned iodine blue-black? What does that tell you is in it?

3. Which powder(s) had a basic pH?
4. Give the chemical formula for each of the four powders.